Equations of Planes: Cartesian Equations

Questions 1–36

Question 1. It is given that the plane Π has equation $2x + 8y - 3z = 4$. Show that:

a) $(3, 1, \frac{10}{3})$ lies on the plane Π

b) $(21, -2, 7)$ does not lie on the plane Π

Question 2. The point A has coordinates $(3, -5, 4)$. Determine which of the following planes contains the point A:

a) $\Pi_1 : 2x + y + 6z = 24$

b) $\Pi_2 : -5x - 2y + 3z = 7$

c) $\Pi_3 : 8x + 3y - 8z = -23$

d) $\Pi_4 : \frac{1}{2}x + \frac{2}{3}y + 3z = 6$

Question 3. It is given that the plane Π has equation $3x - 5y + z = 9$. Show that:

a) contains the vector $\begin{pmatrix} -16 \\ -7 \\ 13 \end{pmatrix}$

b) contains the vector $\begin{pmatrix} 13 \\ 4 \\ -19 \end{pmatrix}$

c) does not contain the vector $\begin{pmatrix} -13 \\ -7 \\ -16 \end{pmatrix}$

Question 4. Write down a cartesian equation of the plane passing through the given point and perpendicular to the given vector:

a) $A(1, 4, 1)$, $\mathbf{n} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$

b) $A(6, 1, -2)$, $\mathbf{n} = \begin{pmatrix} -1 \\ 4 \\ -3 \end{pmatrix}$

c) $A(-3, -4, -9)$, $\mathbf{n} = \begin{pmatrix} 5 \\ -1 \\ -2 \end{pmatrix}$

d) $A(\frac{1}{2}, -\frac{3}{2}, \frac{5}{2})$, $\mathbf{n} = \begin{pmatrix} -2 \\ 3 \\ 10 \end{pmatrix}$

e) $A(3, 0, -2)$, $\mathbf{n} = \begin{pmatrix} 0 \\ 7 \\ -8 \end{pmatrix}$

f) $A(-\frac{1}{3}, \frac{1}{4}, \frac{2}{3})$, $\mathbf{n} = \begin{pmatrix} -4 \\ 15 \\ 0 \end{pmatrix}$

Question 5. The point $A(2,1,3)$ lies on a plane. The vector $\mathbf{n} = \begin{pmatrix} 5 \\ -2 \\ 4 \end{pmatrix}$ is perpendicular to the plane.

a) Find the cartesian equation of the plane.
b) Determine whether the points $B(3,1,-5)$ and $C(-2,0,8)$ lie on the plane.

Question 6. Find the cartesian equation of the plane that is parallel to $4x - y + z + 2 = 0$ and contains the point $(1,1,4)$.

Question 7. Find the cartesian equation of the plane that is parallel to $-3x+8y-z-7=0$ and contains the point $(2, \frac{1}{2}, -8)$.

Question 8. Find the cartesian equation of the plane that is parallel to $kx+3y-2kz-1=0$ and contains the point $(3, -4k, 5)$, where k is a constant.

Question 9. It is given that the plane Π with equation $-x + 6y - 2z = 3$ contains the vector $\mathbf{i} + \mathbf{j} + p\mathbf{k}$.

a) Find the value of p.

b) Hence find the cartesian equation of a plane that is perpendicular to Π and which passes through the point $(2, 1, 1)$.

Question 10. The plane Π_1 has equation $kx - 6y + z = 7$, where k is a constant, and the plane Π_2 has equation $4x + 8y - 3z = -2$. It is given that the angle between the two planes Π_1 and Π_2 is $90°$. Find the value of k.

Question 11. The plane Π_1 has equation $5x + y - kz = 3$ and the plane Π_2 has equation $2kx - 7y + 2z = 9$, where k is a constant. Given that Π_1 and Π_2 are perpendicular, find the value of k.

Question 12. The plane Π_1 has equation $(k+1)x-2y+3kz = 8$ and the plane Π_2 has equation $4kx-(k+2)y-10z = 1$, where k is a constant. Given that Π_1 and Π_2 are perpendicular, find the possible values of k.

Question 13. Find the acute angle, to 1 decimal place, between:

a) the vector $\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ and the plane $x + 4y + 3z = 1$

b) the vector $-\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}$ and the plane $2x - y + 2z = -5$

c) the vector $\mathbf{i} + 8\mathbf{j}$ and the plane $5x - 4z = 2$

d) the vector $9\mathbf{i} - 3\mathbf{k}$ and the plane $-x + 3y + 2z = 25$

Question 14. It is given that the plane Π has equation $x - y - 2z = 3$. Find the acute angle the plane makes with:

a) the XY plane

b) the XZ plane

c) the YZ plane

Question 15. A new road is to be built that goes up a hill. It can be modelled as a plane with equation $2x + 3y - 8z = 4$. The new road connects to a street that can be modelled as a plane with equation $z = 2$. If the angle between the two roads is greater than a 20% incline, then the new road is deemed too steep and is rejected. Determine if the new road will be rejected.

Question 16. Find the acute angle between the following pairs of planes:

a) $\Pi_1 : x + y - z = 2, \quad \Pi_2 : x + y - 2z = 3$

b) $\Pi_1 : 3x + 2y + z = 4, \quad \Pi_2 : 8x + y + 2z = 9$

c) $\Pi_1 : 4x - y - z = 8, \quad \Pi_2 : -x + y + z = -4$

d) $\Pi_1 : -5x - 2y + z + 3 = 0, \quad \Pi_2 : 9x + 10y - 6z - 4 = 0$

e) $\Pi_1 : x - 4z = 9, \quad \Pi_2 : y + 11z = 1$

f) $\Pi_1 : \sqrt{2}x + \sqrt{6}y - z = -10, \quad \Pi_2 : 2\sqrt{2}x + 2\sqrt{3}y - z = 7$

Question 17. Given the equations of the planes $\Pi_1 : x + 12y = 1$ and $\Pi_2 : x + 6y + kz = 2$, where k is a constant, find the range of values of k so that the acute angle between Π_1 and Π_2 is less than $5°$.

Question 18. Given the equations of the planes $\Pi_1 : kx + y + 2z = 3$ and $\Pi_2 : 20y + 39z = 50$, where k is a constant, find the range of values of k so that the acute angle between Π_1 and Π_2 is greater than $1°$.

Question 19. It is given that the plane Π_1 has equation $6x - y + kz = 8$, where k is a positive constant, and that the plane Π_2 has equation $3x - 3y = 5$. It is also given that the angle between the two planes Π_1 and Π_2 is $45°$. Find the exact value of k.

Question 20. It is given that the plane Π_1 has equation $9x + ky - 3z = 2$, where k is a positive constant, and that the plane Π_2 has equation $2x - 2z = -5$. It is also given that the angle between the two planes Π_1 and Π_2 is $30°$. Find the exact value of k.

Question 21. It is given that the plane Π_1 has equation $8x + y + 2kz = -20$, where k is a negative constant, and that the plane Π_2 has equation $4x - 2y = 15$. It is also given that the angle between the two planes Π_1 and Π_2 is $60°$. Find the exact value of k.

Question 22. It is given that the plane Π_1 has equation $8x-4y+z = 5$ and the plane Π_2 has equation $4x+ky-2z = 2$, where k is a constant. It is also given that the acute angle θ between the planes Π_1 and Π_2 is such that $\cos\theta = \frac{1}{9}$. Show that $k = 8 - \frac{4}{\sqrt{3}}$.

Question 23. It is given that the plane Π_1 has normal vector $\mathbf{n}_1 = 10\mathbf{i} - 5\mathbf{j} + p\mathbf{k}$, where p is a negative constant, and that the plane Π_2 has normal vector $\mathbf{n}_2 = 4\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}$. It is also given that:

- the point $(2, 1, -5)$ lies on Π_1
- the angle between the two planes Π_1 and Π_2 is $60°$

Find the cartesian equation of the plane Π_1.

Question 24. It is given that the vector $\mathbf{m} = p\mathbf{i} + q\mathbf{j} + r\mathbf{k}$, where constants $p, q > 0$ and $r < 0$, lies in the plane $3x + z = 3$ and that $|\mathbf{m}| = 4$. It is also given that the angle between $\mathbf{m}$ and the vector $\mathbf{n} = \mathbf{j}$ is $45°$. Find the vector $\mathbf{m}$.

Question 25. It is given that the vector $\mathbf{p} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, where a, b and c are positive constants, lies in the plane $12x - 4y = -3$ and that $|\mathbf{p}| = 6\sqrt{10}$. It is also given that the angle between $\mathbf{p}$ and the vector $\mathbf{q} = \mathbf{k}$ is $30°$. Find the vector $\mathbf{p}$.

Question 26. It is given that the vector $\mathbf{v}$ is a unit vector that is parallel to the plane $x - 2y = 5$. It is also given that $\mathbf{v}$ makes an angle $45°$ with the normal to the plane $8x - 3y + 5z = 1$. Find the two possible vectors $\mathbf{v}$.

Question 27. The points $A(2, 3, -5)$, $B(9, 1, 0)$ and $C(-7, 3, 2)$ all lie in the plane Π.

a) Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

b) Show that the vector $7\mathbf{i} + 47\mathbf{j} + 9\mathbf{k}$ is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

c) Hence find the equation of the plane Π in the form $ax + by + cz = d$, where a, b, c and d are integers.

Question 28. The points $P(-1, 2, 6)$, $Q(3, 2, 8)$ and $R(4, -3, -5)$ all lie in the plane Π.

a) Find $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

b) Show that the vector $5\mathbf{i} + 27\mathbf{j} - 10\mathbf{k}$ is perpendicular to both $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

c) Hence find the equation of the plane Π in the form $ax + by + cz = d$, where a, b, c and d are integers.

Question 29. The points $A(4, 0, -2)$, $B(6, -1, 0)$ and $C(2, 1, -3)$ all lie in the plane Π.

a) Find $\overrightarrow{AB} \times \overrightarrow{AC}$.

b) Find the equation of the plane Π in the form $ax + by + cz = d$, where a, b, c and d are integers.

Question 30. The plane with cartesian equation $ax + by + cz = d$, where a, b, c and d are constants, contains the points $(2, 1, 1)$, $(0, 1, 3)$ and $(1, 0, 4)$. Find the values of a, b, c and d.

Question 31. Find the cartesian equation of the plane that contains the three points:

a) $(1, 0, 1)$, $(1, 1, 1)$ and $(0, 1, 0)$

b) $(2, 1, 1)$, $(-1, 1, 2)$ and $(3, -1, -1)$

c) $(-1, 4, 5)$, $(3, 2, -5)$ and $(4, 2, -3)$

d) $(10, -20, 30)$, $(-20, 40, 50)$ and $(0, 0, -10)$

Question 32. It is given that $A(3, 2, -5)$, $B(4, 1, -2)$, $C(2, 0, 6)$ and D are coplanar. Find the value of k in each of the following cases, where the coordinates of D are:

a) $(k, 1, 2)$

b) $(3, k, 2)$

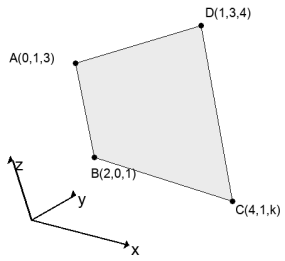
c) $(0, 5, k)$

Question 33. It is given that

$$A = \begin{pmatrix} 2 & -3 & k \\ 4 & 1 & -10 \\ -2 & -11 & 5 \end{pmatrix},$$

where k is a constant. For one particular value of k, the linear transformation associated with the matrix A maps all points to the plane with equation $3x - 2y - z = 0$. Find this value of k.

Question 34. It is given that $A(0, 1, 3)$, $B(2, 0, 1)$, $C(4, 1, k)$ and $D(1, 3, 4)$ all lie on a plane, as shown in the diagram below, where k is a constant.



a) Find the equation of the plane in the form $ax + by + cz = d$, where a, b, c and d are integers.
b) Find the value of the constant k.
c) Find the acute angle the plane makes with the plane with equation $z = 0$.

Question 35. It is given that $A(1,2,-1)$, $B(2,0,-2)$, $C(0,1,1)$ and $D(3,-2,1)$. Additionally, the plane Π_1 contains A, B and C and the plane Π_2 contains A, B and D. Find the acute angle between Π_1 and Π_2.

Question 36. Points $A(0,0,\sqrt{5})$, $B(\frac{\sqrt{15}}{2},0,-\frac{\sqrt{5}}{2})$, $C(-\frac{\sqrt{15}}{2},0,-\frac{\sqrt{5}}{2})$ and $D(0,\sqrt{10},0)$ form a regular tetrahedron.

a) Show that the plane containing A, B and D can be written as $3\sqrt{2}x+\sqrt{3}y+\sqrt{6}z=\sqrt{30}$.

b) Show that the plane containing A, C and D can be written as $3\sqrt{2}x-\sqrt{3}y-\sqrt{6}z=-\sqrt{30}$.

c) Find the acute angle between the ABD and ACD faces of the tetrahedron.